



Koneru Lakshmaiah Education Foundation

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: REAL TIME OPERATING SYSTEMS – (24IN3103HF)

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TITLE OF THE PROJECT	BatteryIQ: Intelligent Battery Management System for Software-Defined Electric Vehicles	
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Abstract:

Electric Vehicles (EVs) are increasingly being adopted as a sustainable alternative to conventional vehicles, with rechargeable battery packs serving as the primary source of energy. The safety, performance, reliability, and lifetime of an EV largely depend on effective battery management. The Battery Management System (BMS) continuously monitors critical battery parameters such as voltage, current, temperature, State of Charge (SOC), and State of Health (SOH) to identify abnormal conditions and maintain safe operation. Conventional BMS mechanisms mainly depend on predefined thresholds and fixed decision strategies, which may have limitations in detecting early battery abnormalities and adapting to changing operating conditions. Furthermore, the emergence of Software-Defined Vehicles (SDVs) requires flexible and intelligent BMS architectures in which battery-management decisions can be dynamically controlled through software. This highlights the need for an intelligent real-time BMS capable of combining embedded systems, Edge-AI, and software-defined decision-making.

To address this need, current research employs machine learning and deep learning models for battery state estimation, fault diagnosis, anomaly detection, and predictive battery management. Existing approaches utilize models such as Long Short-Term Memory (LSTM), Transformer, Autoencoder, Random Forest, and other machine-learning techniques to analyze battery behavior. Building upon these existing models, the proposed project introduces an Intelligent Battery Management System for Software-Defined Electric Vehicles using Embedded C, FreeRTOS, CAN Bus, TensorFlow Lite, Python, IoT, and Docker. The proposed system focuses on a new approach for integrating existing lightweight AI models into a real-time and software-defined BMS through three complementary mechanisms.

The first approach introduces Event-Triggered AI Inference for intelligent battery anomaly detection. Instead of continuously executing the AI model for every battery measurement, the system continuously performs lightweight monitoring of voltage, current, temperature, SOC, and their variations using real-time FreeRTOS tasks. When the monitoring layer identifies suspicious battery behavior, the TensorFlow Lite AI model is triggered for detailed analysis. This approach enables AI-based battery monitoring while reducing unnecessary AI inference and computational overhead.

The second approach introduces a Two-Level Battery Decision System that separates fast battery screening from AI-based verification. The first level continuously performs low-computation parameter and trend analysis to identify potentially abnormal conditions. When suspicious behavior is detected, the second level activates the AI model to verify the condition and determine its severity. The results from both levels are combined to classify the battery condition into states such as normal, warning, or critical. This approach aims to provide timely and reliable battery decisions while maintaining real-time operation.

The third approach introduces a Software-Defined Risk Policy that connects intelligent battery analysis with configurable BMS actions. Instead of directly assigning a fixed response to every detected abnormality, the proposed system converts the detected condition into a battery-risk state and passes it to a software-configurable policy. Based on the selected policy and risk level, the system can generate appropriate actions such as normal operation, warning, charging restriction, vehicle power limitation request, or critical protection. This enables battery-management strategies to be modified through software without changing the underlying sensing hardware, supporting the Software-Defined Vehicle concept.

Collectively, these approaches provide an intelligent and adaptive BMS architecture that combines event-triggered AI inference, two-level battery decision-making, and software-defined risk policies. FreeRTOS provides real-time task scheduling, Embedded C implements the BMS control logic, CAN Bus enables communication with vehicle-side controllers, and TensorFlow Lite supports lightweight Edge-AI inference. Python is used for data preprocessing and model development, while IoT and Docker provide connected monitoring, visualization, and data-management capabilities. The proposed system aims to improve battery anomaly detection, reduce unnecessary computational workload, provide flexible battery-management decisions, and contribute to safer, efficient, and adaptable battery management for next-generation Software-Defined Electric Vehicles.